What is diffusion?

Diffusion occurs when particles move from where there are lots of particles (high concentration) to where there are fewer particles (low concentration). Solids cannot diffuse because their particles are too close together and are strongly attracted to each other. Liquids and gases are able to diffuse. Gases diffuse the quickest because their particles are free to move quickly in different directions. When a liquid or gas is heated, it diffuses more quickly.

☞ **1** Read the following passage and circle the correct word in each pair.

The (movement/shrinking) of particles is called diffusion. A substance spreads out from an area where there is (more/less) of the substance to an area where there is (more/less). Gases diffuse (quickly/slowly) because the particles are moving quickly in all directions. Liquids can also diffuse but the particles move more (slowly/quickly).

Solid particles cannot diffuse because they cannot move, they can only (vibrate/slide over each other). An odour is a (gas/liquid) that is given off by a substance. You can smell perfume because the scent particles diffuse through the air to your nose in the form of a (solid/gas). If you drop a little bit of dye into a jar of water the colour will spread slowly through the water. This is an example of diffusion in (liquids/gases). When particles are (heated/cooled), they diffuse more quickly.

2 List three examples of diffusion.

_____ _____ _____

3 Choose the correct ending for each of the following sentences.

Sentence start	**Sentence endings**		
Diffusion means	contracting.	spreading out.	expanding.
A substance diffuses from an area of high concentration to an area of	the same concentration.	high concentration.	low concentration.
Gases diffuse	quickly.	slowly.	only at very high temperatures.
A dye will diffuse more quickly in	cold water.	room-temperature water.	hot water.

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